Recall Is the Wrong Yardstick: Evaluating Data-Oblivious Vector Quantization on Audio Embeddings

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***Abstract*—Retrieval over audio embeddings underpins query-by-example search, acoustic-scene matching and noise-prototype lookup, and on edge hardware the embedding index—not the model—is usually the memory bottleneck. Data-oblivious vector quantization compresses vectors with no training pass at all, making it attractive for indices built or extended on-device, but it is evaluated almost exclusively by geometric recall against uncompressed nearest neighbours. We evaluate data-oblivious quantization (random rotation followed by per-coordinate optimal scalar quantization) on audio embeddings across three self-supervised speech models—WavLM-Base+, wav2vec2-base and HuBERT-base—using 10 000-vector indices from MS-SNSD and VoiceBank-DEMAND. Our central result is a *dissociation between geometry and meaning under lossy compression*. At $\sim 30\times$ compression the quantizer loses roughly half of the true nearest neighbours, yet retains $\sim 96\,\%$ of downstream speaker- and noise-type-retrieval accuracy on every model tested. Prior work reaches the same conclusion from the *search-approximation* side, showing downstream quality is stable while Recall@k falls [3]; we establish it for the distinct and harsher case where the representation itself is destroyed, on audio, with a measured effect size replicated across three backbones. We also give a mechanistic account of the random rotation: it flattens a spiky *within-vector* coordinate profile (crest factor $10.2 \rightarrow 3.3$ against an isotropic ideal of 3.4). This produced a cross-model prediction, recorded in advance and confirmed. A third model showed the crest factor is a reliable *binary screen* for whether rotation is worth applying, but *not* a predictor of how large the gain will be—we report the weaker claim the evidence supports. Practically, the untrained quantizer reaches $98\,\%$ of OPQ's recall at $1/434$th the index build cost.**

***Index Terms*—vector quantization, audio embeddings, approximate nearest neighbour search, on-device retrieval, self-supervised speech representations**

I. Introduction

Positioning. This work sits at the intersection of two literatures that have not been connected. The speech community has established that self-supervised audio representations are strongly anisotropic and that their layers differ substantially [8], [9], [10]. The vector-quantization community has established that data-oblivious rotation-plus-scalar schemes degrade on anisotropic embeddings [7]. **Neither has asked what the first fact implies for the second**, and neither evaluates audio embeddings under data-oblivious quantization. We do not claim to have discovered either half.

Contributions.

1) **The first evaluation of data-oblivious VQ on audio embeddings**, across three self-supervised models and two corpora. Published benchmarks of these methods cover text, image and ads embeddings and LLM KV caches; audio is untested (Sec. II). Existing audio-embedding compression work uses binary hashing, LSH and PQ—all data-*dependent* or task-trained.
2) **A quantified geometry–meaning dissociation under quantization** (Sec. V-C): $\sim 50\,\%$ neighbour loss costs $\sim 4\,\%$ of downstream task accuracy, replicated on all three models and verified not to be an artifact of the operating point. **This is the paper's central claim**, stated at the scope the evidence supports. We do *not* claim to be first to observe that Recall@k understates deployment quality: [3] show the same for ANN *search* approximation on general corpora and propose $1/\mathrm{Ratio@}k$ in its place. Our contribution is the harder case and the number—search approximation returns *slightly worse true vectors*, while quantization *permanently destroys representation content*, so stability under the former does not imply stability under the latter—measured here on audio, which neither literature covers. That two literatures reach this conclusion by different routes is corroboration, not competition.
3) **A mechanistic account of the random rotation**—the within-vector *crest factor*, a statistic the speech-anisotropy literature does not use (it reports cosine similarity and isotropy)—validated by an advance cross-model prediction (Sec. V-B) and then narrowed, over four successive tests, to a coarse per-layer threshold rather than the quantitative predictor first claimed.
4) **An honest map of what does and does not replicate** (Sec. V-D).
5) **Three practical costs that standard reporting conceals** (Sec. V-F–V-H): the advantage of trained codebooks *grows* with index size on 3/3 models, so results at 10^4 vectors do not transfer to 10^6; the deployable structured rotation loses 0.04–0.065 Recall@10 at matched bits when the dimension is not a power of two, invisible in bits-per-coordinate tables; and layer choice matters comparably for *quantized retrieval*, with the conventional final layer worst on 3/3 models.

II. Related work and positioning

What is already known, and which we do not claim. Three relevant facts are established.

(i) Data-oblivious rotation-plus-scalar quantization degrades on *anisotropic* embeddings: the underlying analysis assumes vectors uniformly distributed on the sphere and does not extend to spaces where a few directions carry most of the variance [7]. That source is an industry technical report rather than peer-reviewed, and we mark it as such wherever it is load-bearing.

(i-b) The data-oblivious regime is not a one-method field. RaBitQ [2] randomly rotates a bi-valued codebook and derives an unbiased distance estimator with a proven error bound, and was the first practically deployable method to attain the asymptotically optimal space–accuracy trade-off. We situate the method under test alongside it rather than presenting the regime as unexplored. Like TurboQuant [1], RaBitQ is evaluated on text and image ANN benchmarks.

(ii) Self-supervised speech representations *are* strongly anisotropic—wav2vec2 embeddings are reported as occupying a narrow cone with layerwise cosine similarities of 0.9–1.0 in upper layers [10], [11], consistent with the $+0.83$ to $+0.93$ we measure at final layers (Sec. V-A).

(iii) Layers of these models differ substantially in what they encode, and layer-wise analysis is an established subfield [8], [9].

We claim none of these. Our anisotropy and layer measurements reproduce known properties *in service of a question neither literature has asked*: what those properties imply for data-oblivious quantization of audio indices.

On evaluation metrics. That Recall@k is a pessimistic proxy is *not* our discovery. [3] show that downstream indicators—classification accuracy, RAG answer quality—stay stable while Recall@k drops substantially in ANN *search*, and propose the inverse approximation ratio $1/\mathrm{Ratio@}k$ instead. Their setting differs from ours in the way that matters: they hold the vectors exact and vary how hard the index is searched, so a missed neighbour is merely a *slightly more distant* true vector. Under quantization the vectors themselves are damaged and the information is gone. Stability under the first does not entail stability under the second, and we test the second. On the compression side, CoRECT [4] benchmarks eight compression families on text retrieval up to 100 M passages and finds the best method varies by model—the closest existing work to Sec. V-F, and likewise audio-free.

What is not covered by prior work. Published benchmarks of these quantizers use arXiv titles, DBpedia entities, Wikipedia, product catalogues, CLIP image and ads embeddings—**no audio**. The audio side is no better covered from its own direction: MAEB [16] evaluates 50+ audio embedding models over 30 tasks but measures **uncompressed** embedding quality only, with no quantization, index-memory or on-device dimension. Existing audio embedding compression is data-dependent or task-trained throughout, e.g. jointly learned embeddings and balanced hash codes for query-by-example [15].

TABLE I
Anisotropy of the three embedding spaces at the final layer. All three occupy 1–3 % of their 768 ambient dimensions.

	WavLM	wav2vec2	HuBERT	isotropic ref.
mean pairwise cosine	$+0.830$	$+0.929$	$+0.853$	-0.000
effective dim.	$2.9\,\%$	$1.4\,\%$	$2.1\,\%$	$92.9\,\%$

A distinction worth stating explicitly. wav2vec2 and related models use product quantization *inside their own training objective* (Gumbel-softmax codebooks over encoder features). That is unrelated to compressing utterance embeddings for a retrieval index, which is our subject.

III. Method

Random rotation (Hadamard or dense Gaussian) followed by a per-coordinate Lloyd–Max quantizer derived from the analytic post-rotation coordinate distribution; norms stored in float32 and counted in the bit budget. Baselines: faiss PQ [5] and OPQ [6] (trained on held-out data), uniform scalar quantization, and the identical quantizer with rotation disabled as the ablation.

Implementation validation. Measured MSE distortion falls below the source method's proven upper bound [1] at every bitrate (b=1: 0.403 vs 0.680; b=4: 0.0094 vs 0.0106), within 2.5–$14.4\,\%$ of its refined values. 34 unit tests cover the quantizers and the retrieval evaluation, including brute-force verification of the top-k and recall routines.

IV. Experimental setup

Indices of 10 000 MS-SNSD utterances with 1 000 disjoint queries; semantic tasks use VoiceBank-DEMAND speaker labels (824 clips) and MS-SNSD noise-type labels (1 100 clips). Embeddings come from pretrained WavLM-Base+ [12], wav2vec2-base [13] and HuBERT-base [14], mean-pooled over time. Ground truth is exhaustive float32 search. PQ/OPQ codebooks are fitted on a held-out 10 000-vector bank, never on the vectors scored. Comparisons use paired Wilcoxon tests and bootstrap CIs over queries.

V. Results

A. Audio embedding spaces are extremely anisotropic

Table I places all three models far from the uniform-on-sphere case the method is optimal for. This confirms a known limitation (Sec. II) in a new domain.

B. Why the rotation works, and when it does not

The prediction that rotation would be near-worthless on wav2vec2 was recorded *before* that benchmark ran, with the explicit condition that equal gains would falsify the mechanism (Fig. 1).

HuBERT was then added to test whether crest factor predicts the *magnitude* of the gain. **It does not**—HuBERT is the spikiest yet gains less than WavLM, inverting the predicted ordering. We therefore report the crest factor as a **binary screen** separating "rotation essential" ($\mathrm{crest} \gg 3.4$) from

TABLE II

THE ROTATION FLATTENS A SPIKY WITHIN-VECTOR COORDINATE PROFILE. ROTATION IS WORTH LITTLE ON THE MODEL WHOSE COORDINATES ARE ALREADY NEAR-ISOTROPIC.

	HuBERT	WavLM	wav2vec2	ideal
crest (raw → rot.)	12.53→3.40	10.22→3.34	3.62→3.29	3.37
kurtosis (raw)	+52.97	+38.75	+0.57	−0.00
rel. R@10 gain	**+40.2 %**	**+49.4 %**	**+2.8 %**	—

TABLE III

AT 1 BIT/COORDINATE (∼30× COMPRESSION), ROUGHLY HALF THE NEIGHBOUR LIST IS DESTROYED WHILE ∼96 % OF DOWNSTREAM TASK ACCURACY SURVIVES, ON EVERY MODEL.

model	Recall@10 kept	speaker P@10 kept	noise P@10 kept
WavLM	48 %	96.5 %	96.0 %
wav2vec2	53 %	95.7 %	96.9 %
HuBERT	70 %	**98.6 %**	**97.3 %**

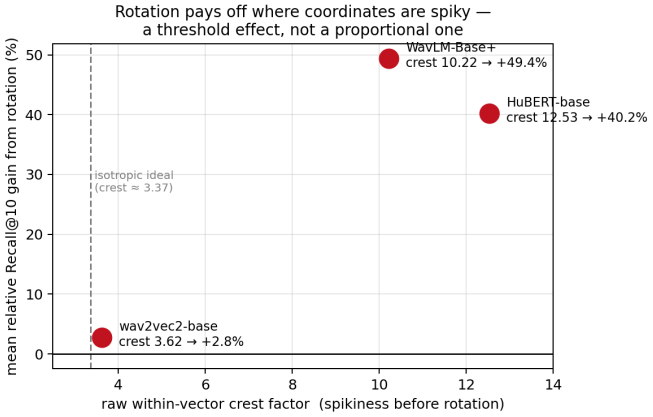


Fig. 1. Rotation benefit against raw within-vector crest factor across the three models. The relationship is a threshold, not a proportionality.

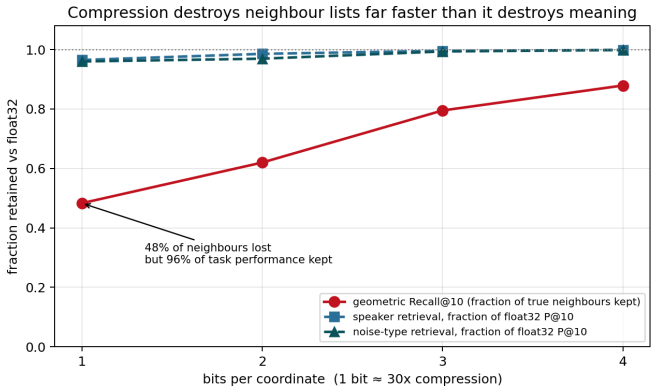


Fig. 2. Recall retention collapses with bit-width while both semantic curves stay near 1.0.

"rotation near-worthless" (crest ≈ 3.4), and explicitly *not* as a quantitative predictor.

C. Geometry and meaning come apart—the central result

What compression removes is largely the fine ordering *among near-ties*, which the downstream task does not depend on (Table III, Fig. 2).

Not an artifact of the operating point. Sec. V-H shows the final layer—used for all results above—is the *worst* of twelve for retrieval, which raises the possibility that the dissociation only appears at a poor operating point. Repeating the measurement on WavLM's best layer (layer 2) shows the effect is if anything **stronger** there: 62.3 % of neighbours retained against **99.7** % of speaker-retrieval and 97.5 % of noise-type-retrieval accuracy. The dissociation is a property of quantization, not of the layer chosen.

D. What replicates, and what does not

The sign-flip deserves comment as a methodological point. Observed first on WavLM, it failed to replicate on wav2vec2; rather than generalise or discard it, we scoped it to spaces where the rotation does substantial work (crest $\gg 3.4$) and noted that wav2vec2 is inert by construction. HuBERT—run afterwards, and the spikiest model of the three—reproduced the WavLM pattern exactly: rotation significantly *helps* speaker retrieval ($p \leq 0.024$ at 2–4 bits) and significantly *hurts* noise-type retrieval ($p \leq 3.3\text{e}{-3}$ at 1–3 bits). The scoped claim was recorded before that test and survived it; the unscoped version would have been contradicted.

The advantage is strongly model-dependent (Table V): the untrained method wins 0/4 budgets on WavLM, 1/4 on wav2vec2 and 2/4 on HuBERT. It also significantly beats *plain PQ* at 3–4 bits on WavLM (-0.029, p=3.6e−12) and at every budget on wav2vec2—OPQ's learned rotation, not product quantization itself, is what makes the data-dependent side competitive.

E. Cost

At ∼3100 bits/vector OPQ buys +0.0199 Recall@10 for **434**× the index build time (360.4 s vs 0.8 s), and requires a training pass that a streaming or on-device index cannot always afford.

F. The advantage of trained codebooks grows with index size

The gap widens with index size on all three models—by up to 3.3× over a single decade (Table VI). This qualifies the practical case directly: our headline comparisons are measured at 10 000 vectors and **should not be assumed to hold at** 10^6. The favourable regime is small on-device indices—the deployment that motivates training-free construction in the first place, and where the gap is smallest on every model (+0.017 to +0.068)—while at server scale a training pass is both more necessary and more affordable. We report this as prominently as the results favouring the method.

The penalty's magnitude is also strongly model-dependent (WavLM +0.224 vs HuBERT +0.089 at 10k), so for this deployment the choice of embedding model matters as much as the choice of quantizer.

TABLE IV
REPLICATION STATUS OF EACH FINDING ACROSS THE THREE MODELS.

finding	replication
geometry–meaning dissociation	**3/3 models**
data-dependent advantage widens as budget shrinks	**2/3 cleanly** (HuBERT non-monotone, $\rho = -0.60$)
rotation's task-dependent sign-flip, *as scoped to spiky spaces*	**2/2 spiky models**; inert on the flat model, as predicted

TABLE V
MATCHED-BUDGET RECALL@10, BEST DATA-DEPENDENT MINUS DATA-OBLIVIOUS. POSITIVE FAVOURS THE TRAINED CODEBOOK.

bits/vec	WavLM	wav2vec2	HuBERT
800	$+0.243$	$+0.202$	$+0.040$
1568	$+0.224$	$+0.092$	$+0.089$
2336	$+0.052$	$+0.043$	-0.037
3104	$+0.020$	-0.049	-0.012

G. The deployable structured rotation is not free

Our benchmarks use a dense Gaussian rotation; the variant one would deploy is the structured Hadamard transform. Because $d = 768$ is not a power of two, the Hadamard path pads to 1024 and spends $\mathbf{33}\,\%$ more bits—which a bits-per-*coordinate* table hides. At matched bits per *vector* the ordering reverses, costing 0.04–0.065 Recall@10 on both models tested. Any deployment at a dimension that is not a power of two pays this, and we note the reporting convention that conceals it: **bits per vector, not bits per coordinate, is the comparable unit.**

Implementation caveat: our Hadamard path applies the full matrix as a dense matmul, so its measured encode times reflect an $O(d^2)$ implementation and support no claim about the $O(d \log d)$ fast Walsh–Hadamard transform. The recall comparison is unaffected, depending only on the transform's mathematical effect.

H. Layer choice rivals quantizer choice

All results above use the final encoder layer, the conventional default. Sweeping all 12 WavLM layers (3 000-vector index, 2 bit/coord) shows this is the **worst** layer of the twelve for retrieval: layer 2 attains R@10 $= 0.7980$ against 0.6435 at the final layer—$+\mathbf{0.155}$ absolute, $+24\,\%$ relative, at zero cost. For scale, the best *quantizer* choice at a matched budget is worth about $+0.22$. Layer selection is therefore of comparable value to quantizer selection and is free; practitioners should tune it first.

This replicates on HuBERT, where layer 0 is best (0.8383 vs 0.7802, $+7.4\,\%$). Both models penalise the conventional default, but the optimal layer differs (2 vs 0) and so does the magnitude, so the actionable advice is to *sweep the layers* rather than adopt a fixed one. The sweep is cheap: a single forward pass returns every layer.

Two further consequences. Our absolute recall figures, all measured at the final layer, are a **lower bound** on what this pipeline achieves (method *comparisons* are unaffected, since every method saw identical embeddings). And the within-vector crest factor varies $\mathbf{2.8}\times$ across layers of one model ($5.54 \to 15.57$), a spread comparable to the range across the three models we study—so crest factor is a property of a *layer*, not of a model.

TABLE VI
GAP TO OPQ AT A MATCHED BUDGET (2 BIT/COORD VS OPQ m=192) AS THE INDEX GROWS. THE GAP WIDENS ON ALL THREE MODELS.

index size	WavLM	wav2vec2	HuBERT
1 000	$+0.068$	$+\mathbf{0.017}$	$+0.036$
2 500	$+0.156$	$+0.055$	$+0.051$
5 000	$+0.224$	$+0.102$	$+0.074$
10 000	$+0.224$	$+0.092$	$+0.089$

A correction this sweep forces. wav2vec2's *final* layer has crest 3.62—the "flat" case of Sec. V-B—but its layers 1–8 reach crest 17.15, as spiky as anything in WavLM or HuBERT, and there the rotation is worth $+30$–$40\,\%$. wav2vec2 is therefore not a flat *model*; it has a flat *final layer*. The Sec. V-B prediction was correct as measured (it compared final layers), but its explanation attributed a layer property to a model, and we correct that here.

This also resolves the within-model correlations, which at first appear contradictory ($+0.35$ WavLM, -0.14 HuBERT, $+0.74$ wav2vec2). wav2vec2 is the only model whose layers straddle the flat region ($3.62 \to 17.15$) and the only one with a strong correlation; HuBERT's layers are *all* well above it ($9.38 \to 19.47$), so crest carries no signal there even though rotation helps every layer. The behaviour is a threshold, not a proportionality.

We therefore state the surviving claim precisely: **crest factor is a property of a specific layer, not of a model, and works only as a coarse threshold test.** Screen the layer you intend to deploy.

VI. LIMITATIONS

Three embedding models, one language, English read/near-field speech only. All headline results use the final encoder layer, which Sec. V-H shows is the worst of twelve for retrieval on WavLM; absolute recall figures are therefore a lower bound, though method comparisons are unaffected because every method was evaluated on identical embeddings. Indices are 10 000 vectors, and the scale sweep (Sec. V-F) shows the data-dependent advantage *growing* over the reachable 1k–10k range, so results here should not be assumed to hold at 10^6; the trend direction is measured, its continuation is not. Build-cost timings come from one CPU and were not measured on an idle machine. Two semantic label sets from two corpora; broader downstream tasks untested. There is **no on-device measurement**—the edge motivation is argued from index size and build cost, not demonstrated on edge hardware.

Several ingredients are prior work (Sec. II) and we claim only their combination: the anisotropy limitation of data-oblivious VQ, the anisotropy of speech representations, layer-wise variation in those representations, and audio-embedding

compression via hashing/PQ are each established independently. Our contribution is the connection and its consequences, not any single component.

A citation pass was completed against the quantization, ANN-evaluation, speech-representation and audio-retrieval literatures; one database-level sweep (DBLP / ACL Anthology / ISCA Archive) remains before submission.

VII. Conclusion

Data-oblivious vector quantization loses to trained codebooks on audio embeddings by the standard recall benchmark, for a reason already documented in the literature and here measured on audio for the first time. But the standard benchmark is the wrong yardstick for the applications these indices serve: half the neighbour list can be destroyed while $\sim 96\,\%$ of downstream task accuracy survives, on every model we tested. Once that is measured, and index build cost is counted, the untrained method is competitive or better in exactly the setting that motivated it.

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